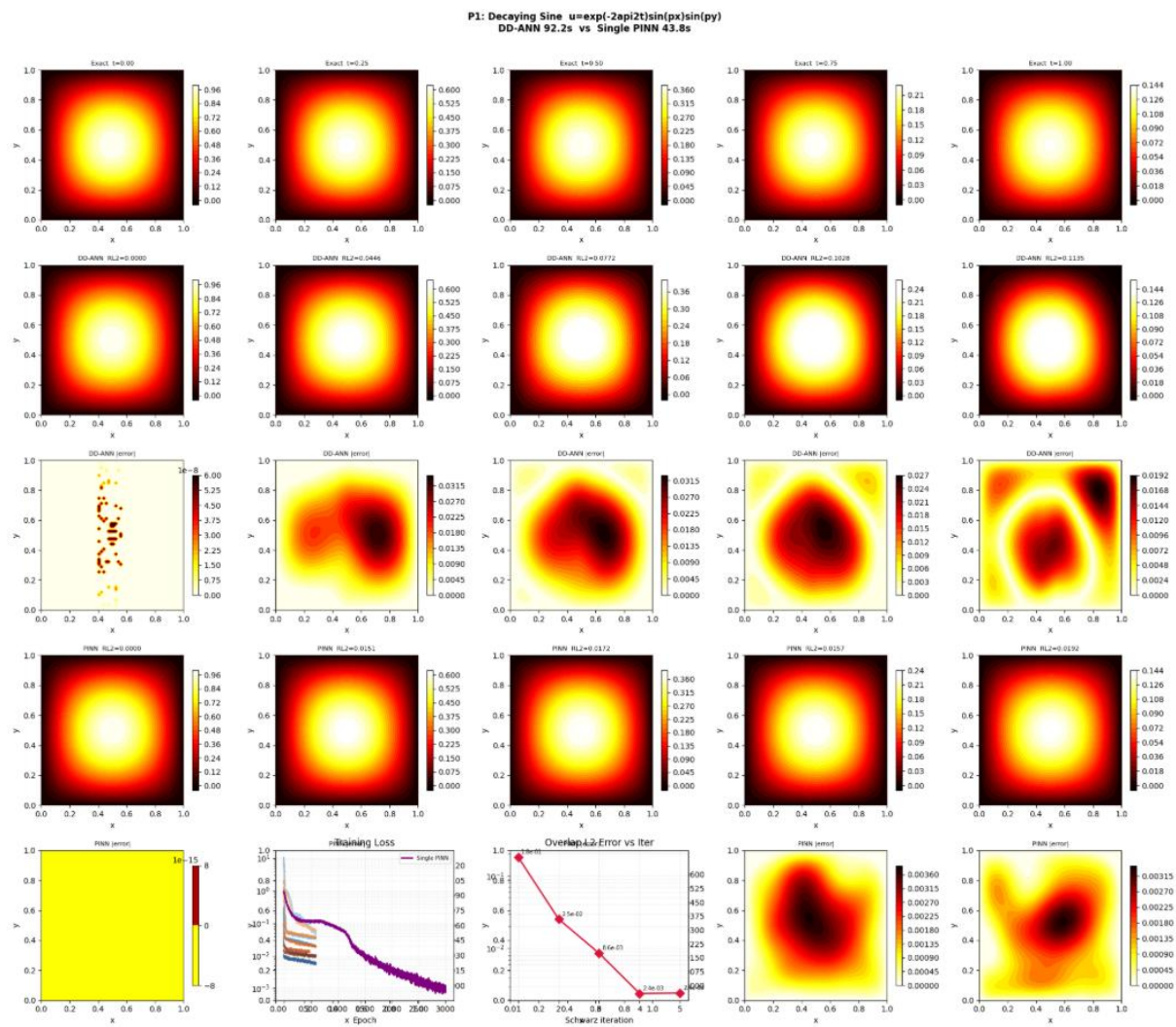


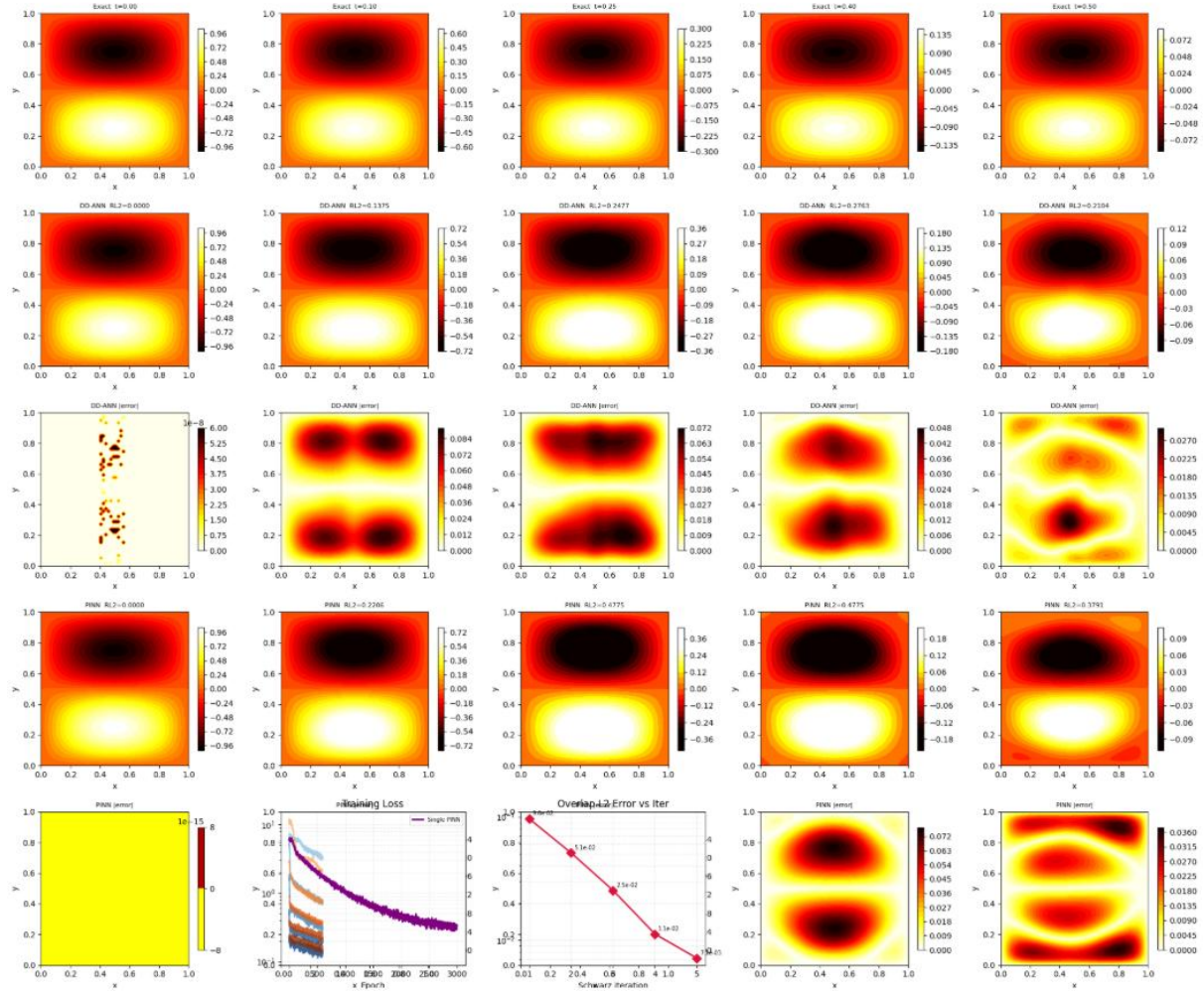
# DDANN-2D (compact hard BC architecture)

## Three Test Problems

| # | Exact $u(x, y, t)$                                                                               | $\alpha$ | $T$ |
|---|--------------------------------------------------------------------------------------------------|----------|-----|
| 1 | $e^{-2\alpha\pi^2 t} \sin(\pi x) \sin(\pi y)$                                                    | 0.1      | 1.0 |
| 2 | $e^{-5\alpha\pi^2 t} \sin(\pi x) \sin(2\pi y)$                                                   | 0.1      | 0.5 |
| 3 | $e^{-2\alpha\pi^2 t} \sin(\pi x) \sin(\pi y) + 0.5e^{-8\alpha\pi^2 t} \sin(2\pi x) \sin(2\pi y)$ | 0.1      | 0.5 |



P2: Higher Mode  $u = \exp(-5\pi i 2t) \sin(\pi x) \sin(2\pi y)$   
 DD-ANN 90.2s vs Single PINN 48.3s



P3: Two-Mode Superposition  
DD-ANN 86.6s vs Single PINN 15.5s

